Patrick Murray

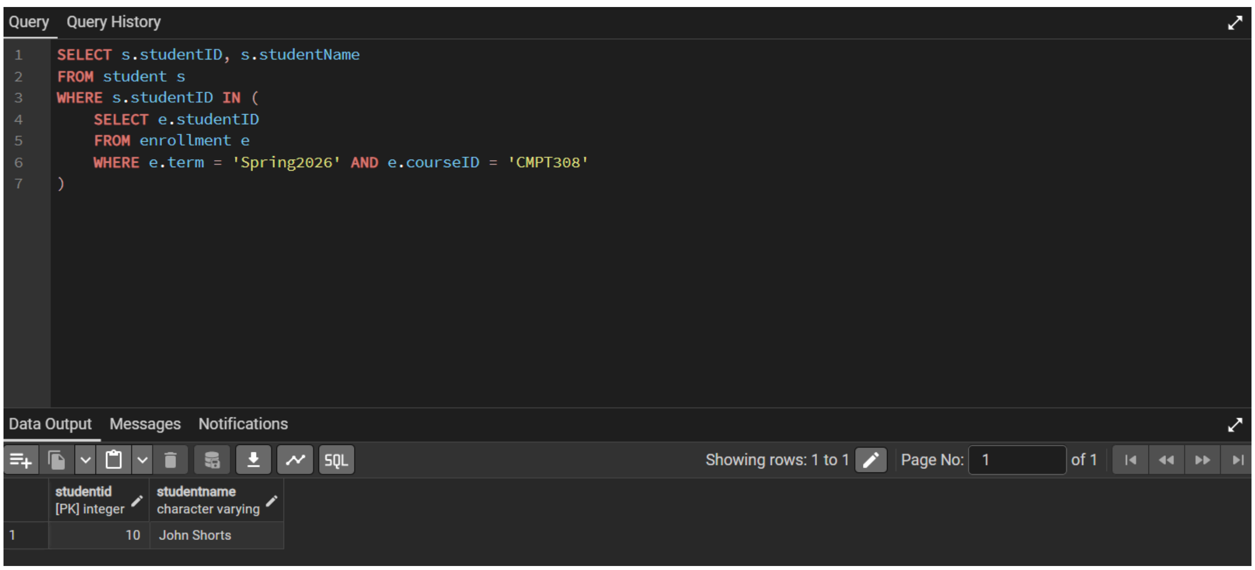
Professor Rivas

Database Management

24 March 2026

Part A:

1:



2:

```sql
SELECT c.courseID, c.title
FROM course c
WHERE EXISTS (
    SELECT 1
    FROM enrollment e
    WHERE e.courseID = c.courseID AND e.term = 'Spring2026'
)
```

Showing rows: 1 to 7

	courseid [PK] character varying	title character varying
1	HIST101	World History
2	CMPT308	Database Manageme...
3	CMPT306	Database Systems
4	CHEM302	Organic Chemistry
5	ECO309	Microeconomics
6	CYBR210	Cyber Security
7	PHY230	Physics I

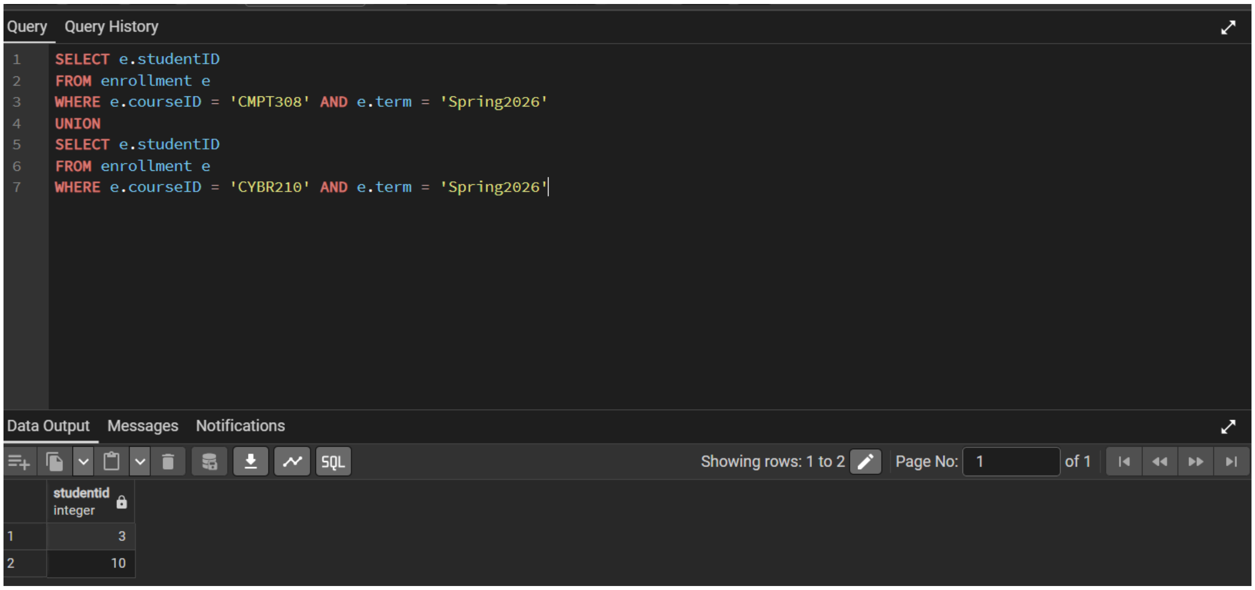
3:

```sql
SELECT s.studentID, s.studentName
FROM student s
WHERE NOT EXISTS (
    SELECT 1
    FROM enrollment e
    WHERE e.studentID = s.studentID AND e.term = 'Spring2026'
)
```

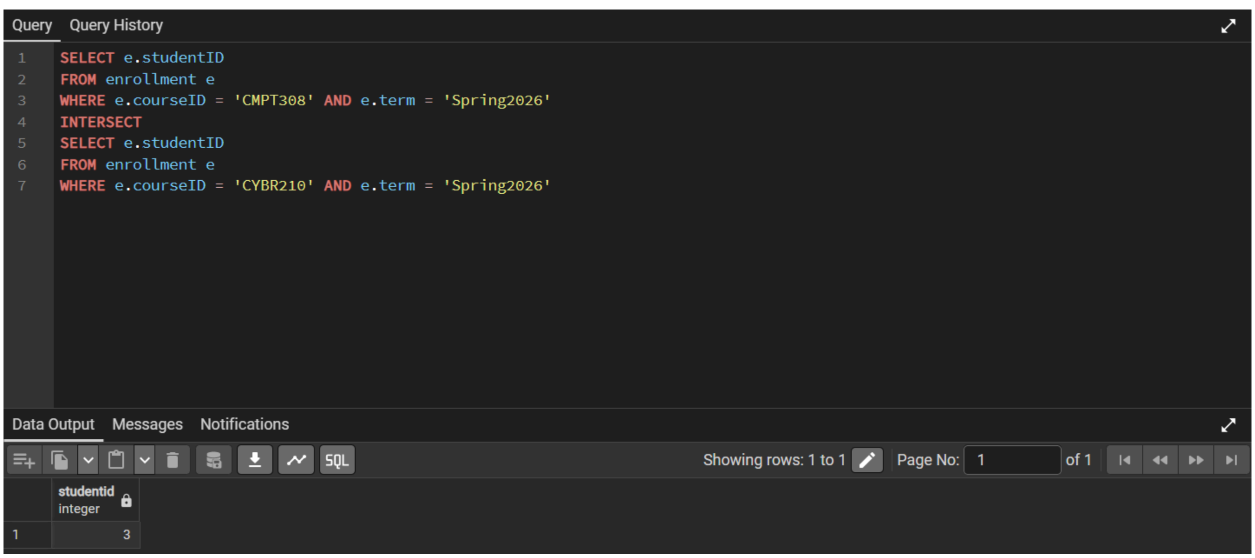
Showing rows: 1 to 2

	studentid [PK] integer	studentname character varying
1	1	Alice Johnson
2	7	Grace Kim

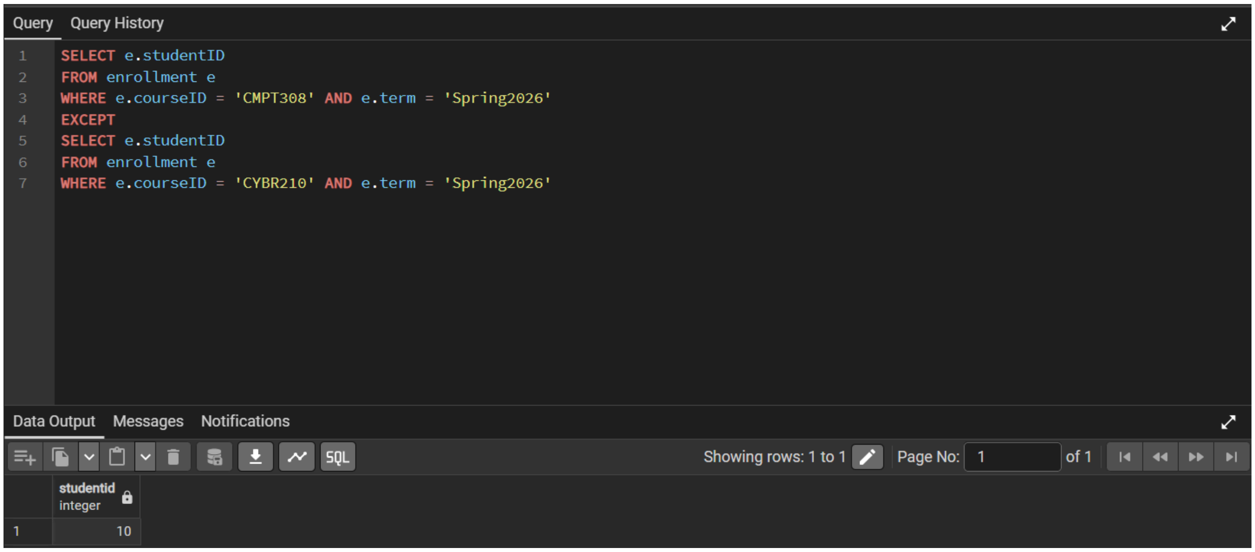
4:


Query Query History
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
UNION
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'
Data Output Messages Notifications
Showing rows: 1 to 2
Page No: 1 of 1
studentid integer
1 3
2 10

5:


Query Query History
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
INTERSECT
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'
Data Output Messages Notifications
Showing rows: 1 to 1
Page No: 1 of 1
studentid integer
1 3

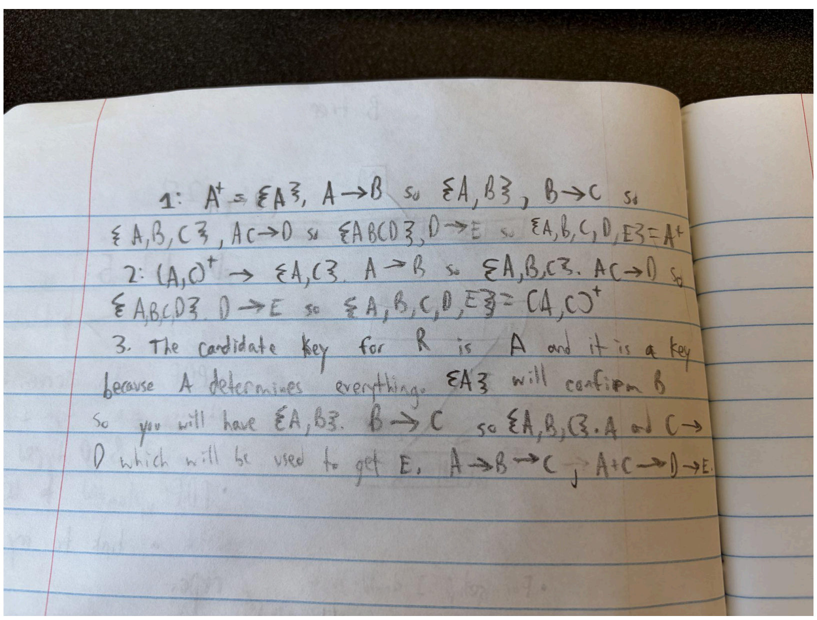
6:



```sql
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
EXCEPT
SELECT e.studentID
FROM enrollment e
WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'
```


	studentid integer
1	10

Part B:



1: $A^+ = \{A\}$, $A \to B$ so $\{A, B\}$, $B \to C$ so $\{A, B, C\}$, $AC \to D$ so $\{ABCD\}$, $D \to E$ so $\{A, B, C, D, E\} = A^+$

2: $(A, C)^+ \to \{A, C\}$. $A \to B$ so $\{A, B, C\}$. $AC \to D$ so $\{A, B, C, D\}$. $D \to E$ so $\{A, B, C, D, E\} = (A, C)^+$

3. The candidate key for R is A and it is a key because A determines everything. $\{A\}$ will confirm B so you will have $\{A, B\}$. $B \to C$ so $\{A, B, C\}$. A and $C \to D$ which will be used to get E. $A \to B \to C$, $A + C \to D \to E$.

Part C:

C 1: Enrollment wide is in 1NF because all of its attributes are atomic and will have no more than 1 value for 1 enrollment. For example, a student will not have a list of grades for 1 enrollment.

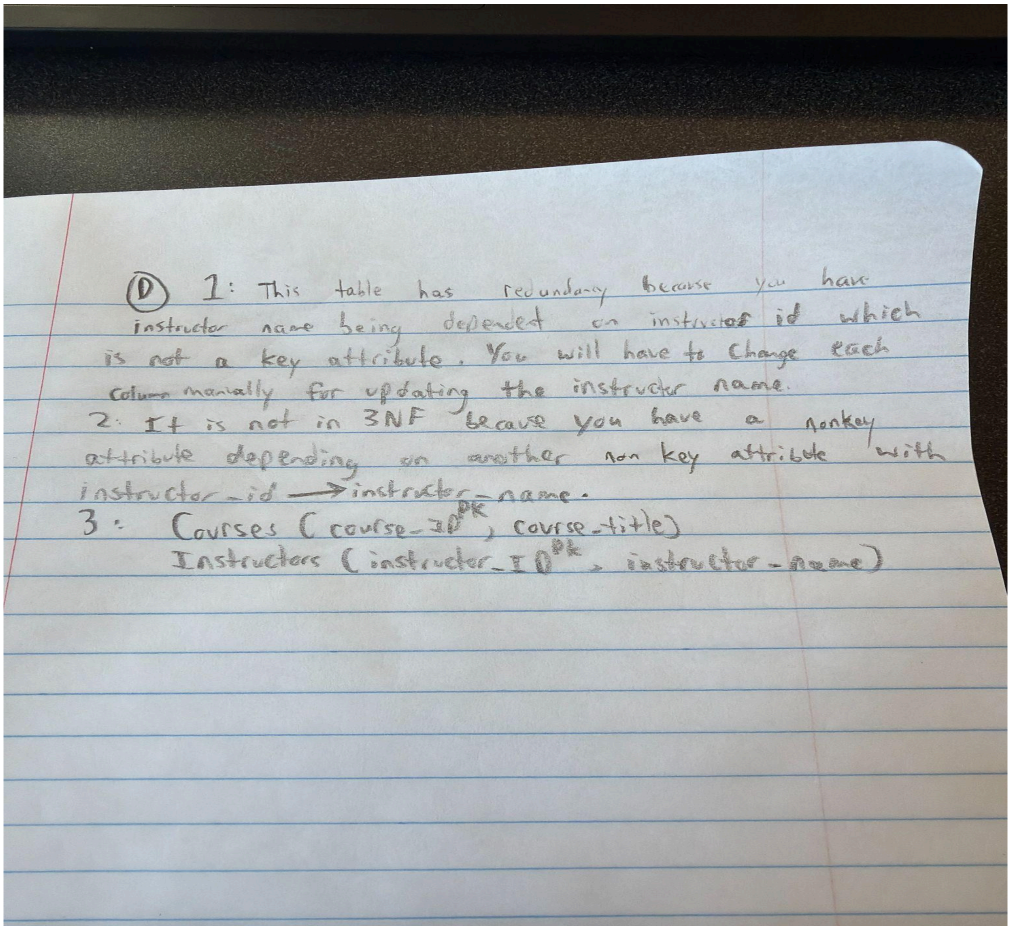
2: EnrollmentWide does violate 2NF since you have 2 partial dependencies with StudentId → studentName and course_id → courseTitle.

3: Students (student_ID PK , student_Name)

(courses (course_Id PK , course_title)

Enrollments (student-ID PK,FK , course-ID PK,FK , term PK , grade)

Part D:



D 1: This table has redundancy because you have instructor name being depended on instructor id which is not a key attribute. You will have to Change each column manually for updating the instructor name.

2: It is not in 3NF because you have a nonkey attribute depending on another non key attribute with instructor_id ⟶ instructor_name.

3: Courses (course_ID^{PK}, course_title)

Instructors (instructor_ID^{PK}, instructor_name)

GITHUB Link: [DatabaseLabs/Lab5 at main · Not-Patrick1/DatabaseLabs]